

1. Wakefulness

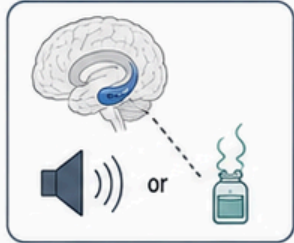
1 Learning task



2 Memory encoding



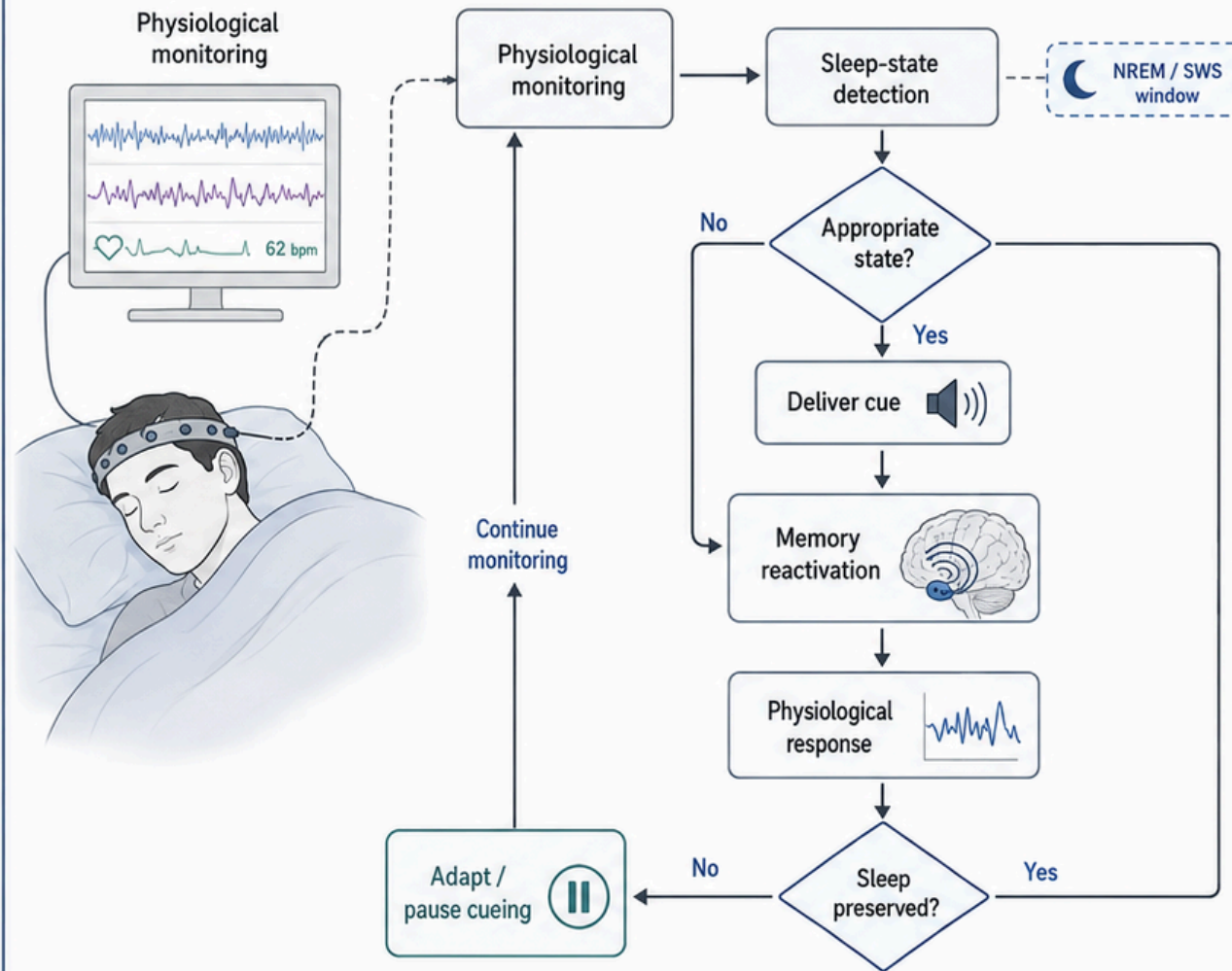
3 Cue association



Associated cue

Memory and cue are linked before sleep.

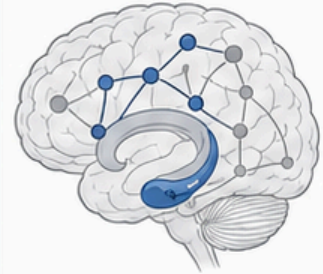
2. Closed-loop sleep



State-dependent, adaptive cueing maximizes reactivation while preserving sleep physiology.

3. Outcome

1 Selective consolidation



2 Improved recall



Strengthened cue-linked memories and better recall.



Cue-linked memories are reactivated during appropriate sleep states while preserving sleep continuity.